SOFTWARE ENGINEERING PROJECT

Termtem

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Abstract

TermTem is an application designed to reduce communication barriers between people with hearing disabilities and hearing individuals in Thailand. Communication difficulties often occur because many Thai people do not understand Thai Sign Language, making everyday interactions challenging for members of the deaf community. The objective of this project is to develop a system that facilitates communication through multiple interaction methods, including sign language recognition, speech recognition, and conversational AI.

The proposed system integrates several machine learning components such as sign language recognition, speech recognition, natural language understanding, and text-to-animation generation. These technologies allow the system to interpret sign language gestures from video input, convert speech into text, and generate responses through speech or sign language animations (Avatar).

TermTem aims to provide near real time communication assistance for hearing disabilities users, individuals learning Thai sign language, and general users who need to communicate with deaf individuals. By applying artificial intelligence technologies, this project seeks to improve accessibility, support communication, and enhance the quality of life for people with hearing disabilities in Thailand.

Acknowledgement

Put your acknowledgement paragraph here.

Sorasit Kateratorn
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Table of Contents

List of Tables

List of Figures

Chapter 1

Introduction

1.1 Background

Communication is an essential part of daily life, yet people with hearing disabilities often face significant challenges when interacting with others. Many members of the deaf community rely on sign language as their primary form of communication, but most hearing people do not understand it. This gap creates communication barriers in situations such as education, healthcare, employment, and emergency services. In Thailand, where Thai Sign Language is widely used, these challenges are even more evident due to the limited availability of professional interpreters, especially in urgent or everyday scenarios.

Although modern technologies such as artificial intelligence, speech recognition, and computer vision have been developed to support communication, existing solutions still have limitations. Many systems focus mainly on direct or word by word translation, which may not accurately capture the meaning, context, or natural flow of communication. Additionally, some solutions are not designed for real-time interaction or are not user friendly for everyday use.

Therefore, there is a need for a more effective communication system that can provide natural and continuous interpretation rather than simple word by word translation. Such a system should support smooth interaction between hearing disabilities individuals and hearing users while maintaining context and meaning during communication.

This project proposes TermTem, an AI-based communication assistant that integrates sign language recognition, speech recognition, conversational AI, and real time animation displayed through avatar to support

real time communication between hearing disability individuals and hearing users.

1.2 Problem Statement

People with hearing disabilities often experience communication difficulties when interacting with individuals who do not understand Thai Sign Language. In many real life situations such as hospitals, workplaces, public services, or emergency cases, communication becomes difficult because sign language interpreters are not always available. Existing solutions mainly rely on human interpreters or limited translation tools, which may be costly, unavailable, or not suitable for real time communication.

In addition, many current technological solutions focus on word by word translation, which may not accurately capture the meaning, context or natural flow of interaction between users.

Therefore, there is a need for an intelligent system that can assist in communication. Such a system should be able to translate sign language, speech, and text in real time to reduce communication barriers and improve accessibility for people with hearing disabilities.

1.3 Solution Overview

This project proposes TermTem, an AI-based communication assistant that integrates sign language recognition, speech recognition, conversational AI, and real time animation displayed through avatar to support real time communication between hearing disability individuals and hearing users.

1.3.1 Features

1. Speech to Sign Translation with Avatar: The system converts speech or text input from hearing users into Thai Sign Language animations displayed through an avatar, enabling hearing disabilities users to understand messages in a natural and visual form.
2. Speech Recognition: The system captures and converts spoken language from hearing users into text for further processing by the system.
3. Sign Language Animation Generation: The system generates accurate and continuous sign language animations using an avatar to represent gestures clearly and naturally, focusing on smooth communication rather than word by word translation.
4. Conversational AI Assistant: The system uses a conversational AI model to understand user intent and generate meaningful responses, improving the quality and flow of communication.

1.3.2 Extra Features

1. Learning Module: The system provides learning support through interactive quizzes to help users learn and improve their Thai Sign Language skills.

1.4 Target User

- **People with hearing disabilities**
  - Demographics: Individuals of all ages in Thailand who primarily use Thai Sign Language as their main form of communication.
  - Skill level: Basic smartphone proficiency.
  - Industry or Domain: Daily communication scenarios such as healthcare services, public services, workplaces, and emergency situations.
- **Learners of Thai Sign Language**
  - Demographics: Students, educators, family members, or individuals interested in learning Thai Sign Language.
  - Skill level: Beginners in sign language who require clear visual demonstrations and accurate sign language animations for learning.
  - Industry or Domain: Education.

1.5 Terminology

Term	**Description**
Thai Sign Language	The primary sign language used by the deaf community in Thailand.
Sign Language Recognition	A computer vision technique used to detect and interpret sign language gestures.
Speech Recognition	Technology that converts spoken language into text.
Text-to-Speech	Technology that converts text into spoken audio.
Conversational AI	AI technology that enables systems to understand and respond to human language.
Avatar Animation	A digital character used to visually represent sign language gestures.

Chapter 2

Literature Review and Related Work

2.1 Competitor Analysis

COMPARISON TABLE

Name	AI	Avatar	Thai Language Supported	Educational	Interpreter
TermTem	✓	✓	✓	✓	✓
NECTEC TTRS	✗	✗	✓	✗	✓
Development of Web Application to Detect ThSL Alphabet via Hand Gesture (2024)	✓	✗	✓	✗	✓
KARA Technology	✓	✓	✗	✓	✓
MySL Web Application	✓	✗	✓	✓	✗

Figure 2.1: Comparison Table of TermTem and existing competitors

2.2 Literature Review

2.2.1 NECTEC TTRS System

The Telecommunication Relay Service (TTRS) developed by NECTEC is a system designed to help hearing-impaired users communicate with hearing individuals. The system provides services such as remote sign language interpretation, emergency communication, and video-based interaction.

However, the system relies heavily on human interpreters and does not fully utilize AI or automated avatar based sign language generation. This limits scalability and real time usability in everyday communication.

This work informs our system design by highlighting the importance of supporting communication between hearing and deaf users. However, TermTem improves upon this by introducing AI-based automation and avatar driven sign language representation.

2.2.2 Thai Sign Language Recognition Research (2024)

The research on Thai Sign Language alphabet detection using hand gestures focuses on applying machine learning techniques to recognize hand signs. The system demonstrates that AI can effectively classify sign language gestures based on visual input.

However, this approach mainly focuses on recognition and does not provide a complete communication system. It lacks features such as avatar visualization and interactive communication support.

This research supports our decision to use AI for gesture recognition, but TermTem extends this idea by providing a full communication pipeline from hearing users to deaf users through animated sign language avatars.

2.2.3 KARA Technology

KARA Technology provides AI-based solutions that include avatar systems and communication tools for accessibility. The system demon-

strates the potential of using avatars to enhance human-computer interaction.

However, the platform does not specifically support Thai Sign Language, which limits its usability for Thai users.

This motivates our system to focus specifically on Thai language support while integrating avatar-based communication to improve accessibility.

2.2.4 MySL Web Application (Senior Kasetsart University Project)

The MySL Web Application project focuses on sign language learning and communication support. It provides AI-based features and educational tools to help users understand sign language.

However, the system lacks real time interpreter functionality and avatar based communication, which limits its effectiveness in real world communication scenarios.

This project influences our design by incorporating educational features, but TermTem enhances it by combining learning, communication, and AI-based translation into a single platform.

Chapter 3

Requirement Analysis

3.1 Stakeholder Analysis

Stakeholder	**Role**
Users with hearing disabilities	Primary users who rely on the system for communication from hearing users through sign language animations.
General hearing users	Individuals who want to communicate with hearing disabilities users using speech or text, which will be translated into sign language animations.
Learners of Thai Sign Language	Users who want to learn Thai Sign Language through visual demonstrations and interactive features such as quizzes.
System developers	Responsible for designing and implementing the system.
Organizations staff member	Hospitals, schools, or agencies that may use the system.

3.2 User Stories and Use Cases

- As a general hearing user, I want my speech to be converted into sign language animation through an avatar so that hearing disabilities users can understand me clearly.

- As a general hearing user, I want to type text and have it translated into sign language animation so that I can communicate in situations where speaking is not possible.
- As a user with hearing disabilities, I want to receive information in sign language animation so that I can understand messages naturally without relying on text.
- As a learner of Thai Sign Language, I want to view correct sign language animations for specific words or sentences so that I can learn proper gestures step by step.
- As a learner of Thai Sign Language, I want to take quizzes or practice exercises based on sign language animations so that I can test and improve my understanding of sign language.
- As an organization staff member, I want to communicate with hearing disabilities users using the system so that I can provide services effectively.

3.3 Use Case Diagrams

3.4 User Interface Design

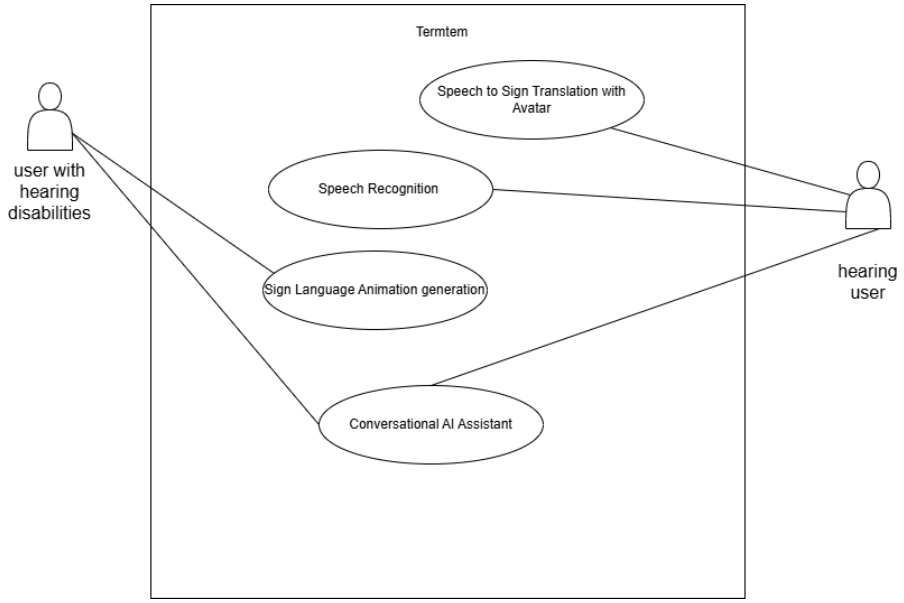

Figure 3.1: Use Case Diagram Design

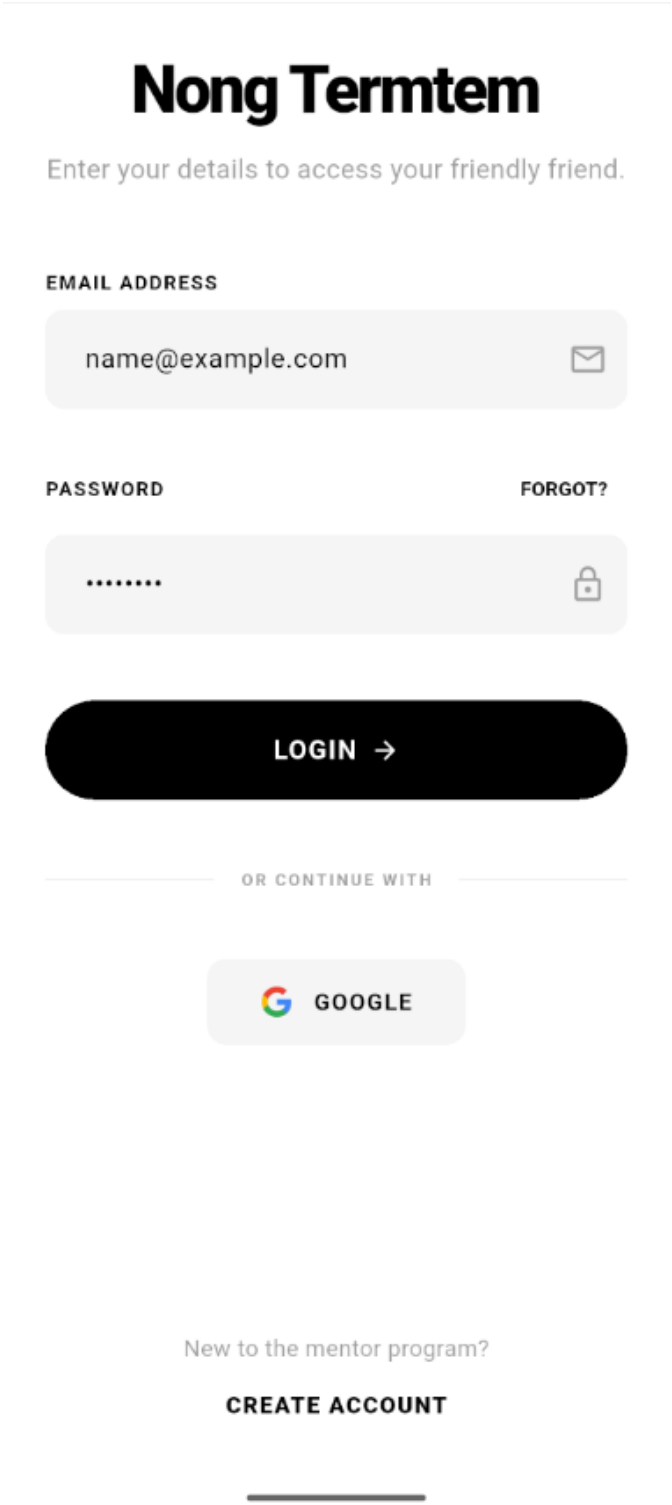

Figure 3.2: Login Page Design

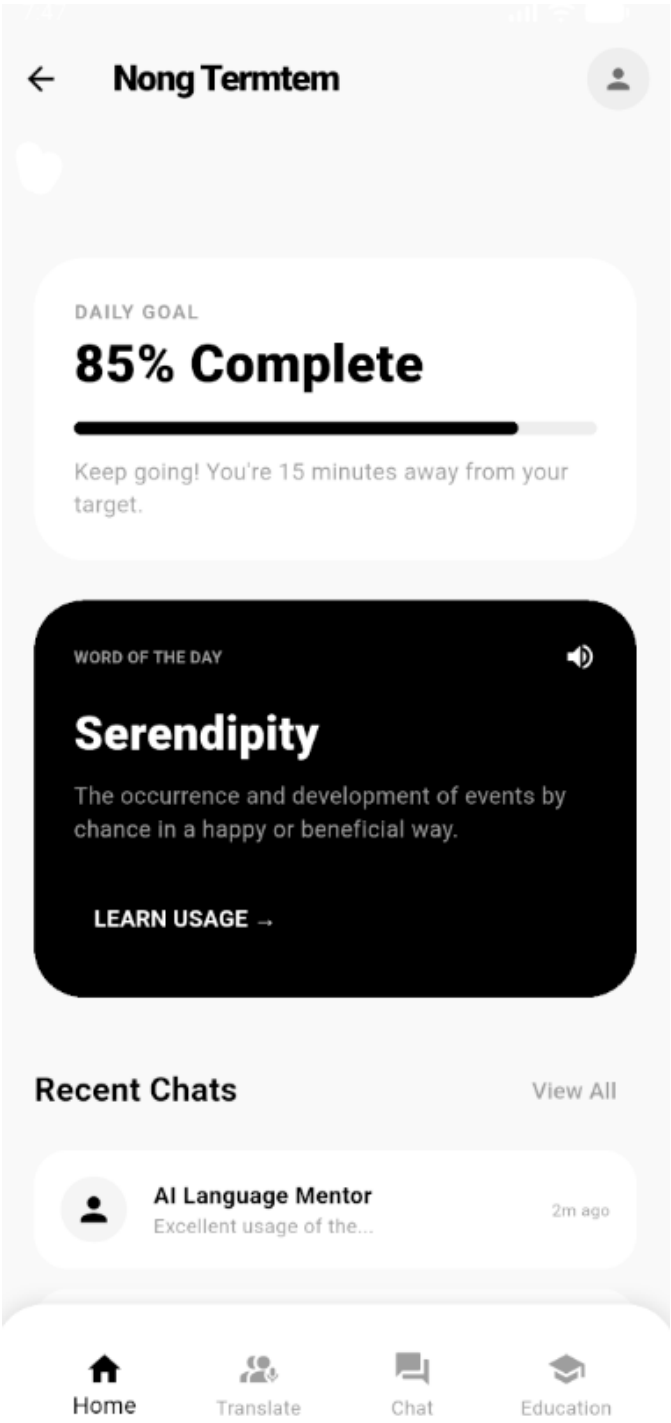

Figure 3.3: Home Page Design

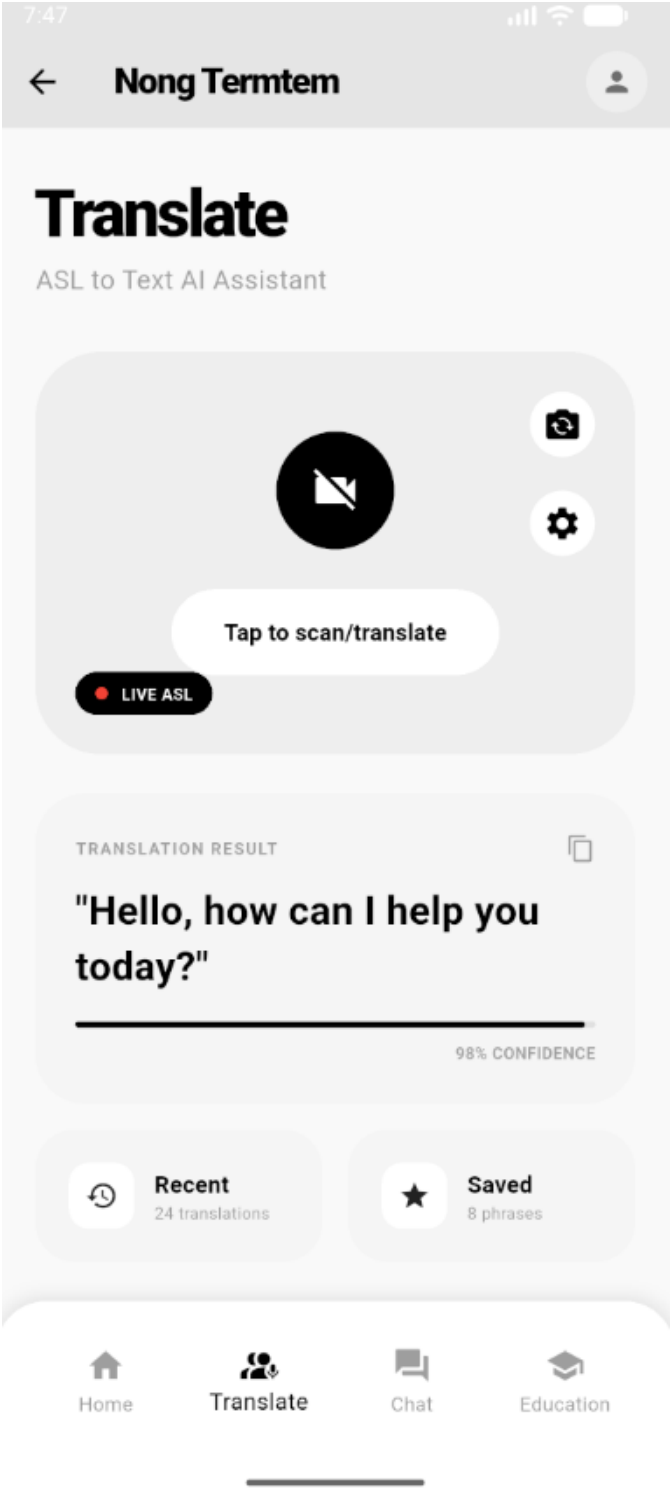

Figure 3.4: Translate Page Design

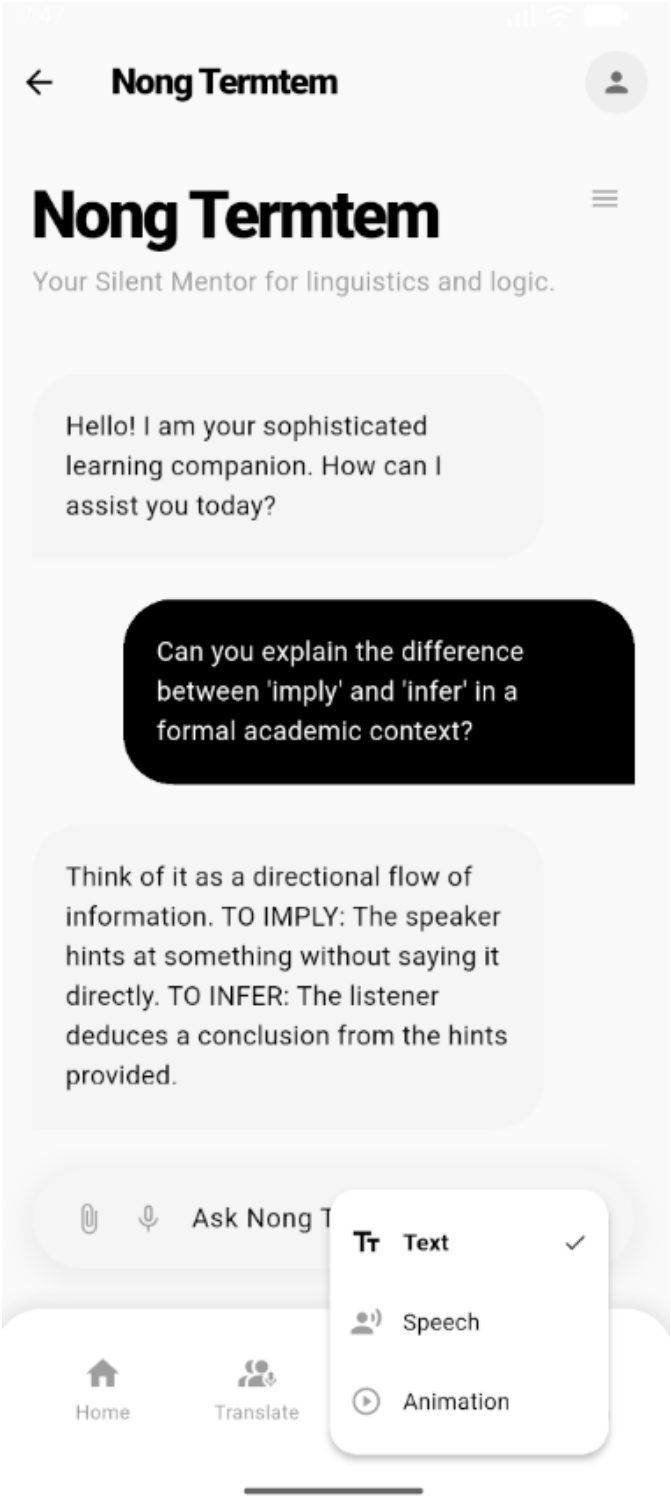

Figure 3.5: Chat Page Design

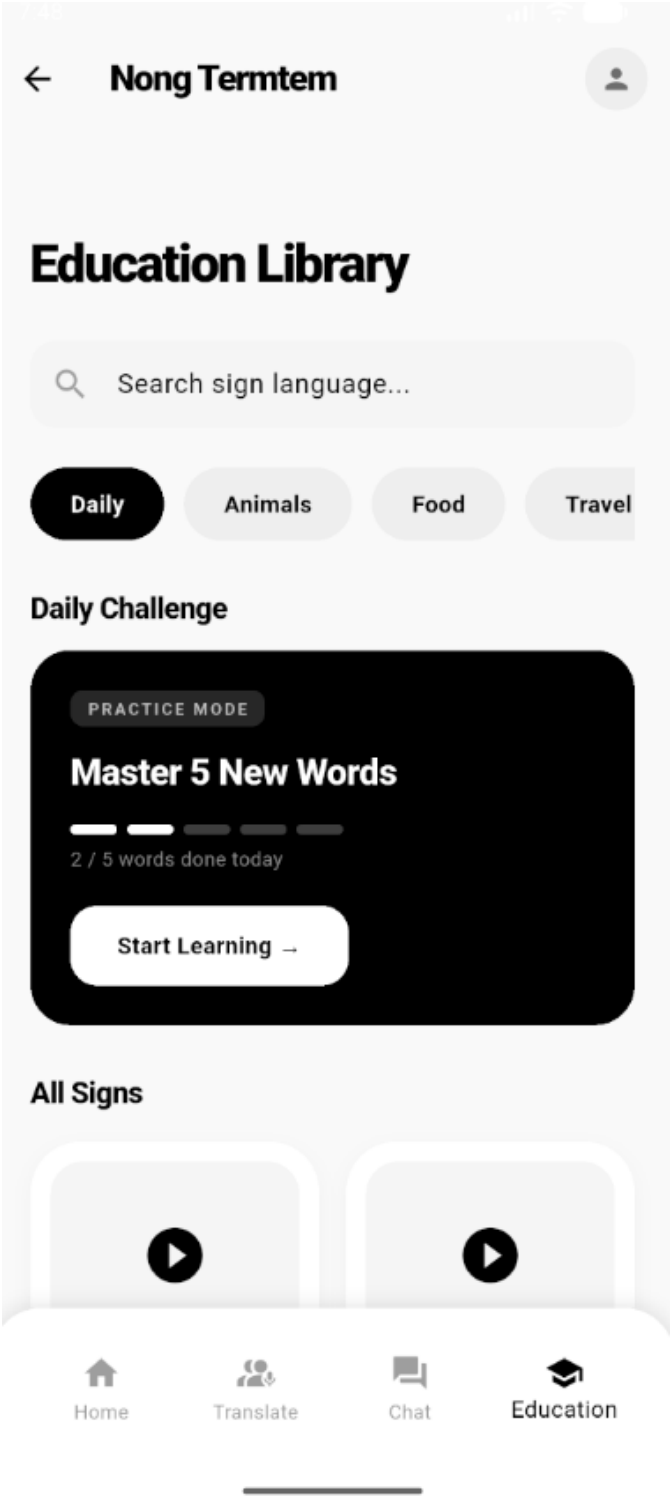

Figure 3.6: Education Page Design

Chapter 4

Software Architecture Design

4.1 System Architecture Overview

The system is designed as a multi-modal AI assistant architecture that integrates mobile application interfaces, AI processing pipelines, and backend services to support communication for users with hearing and speech impairments.

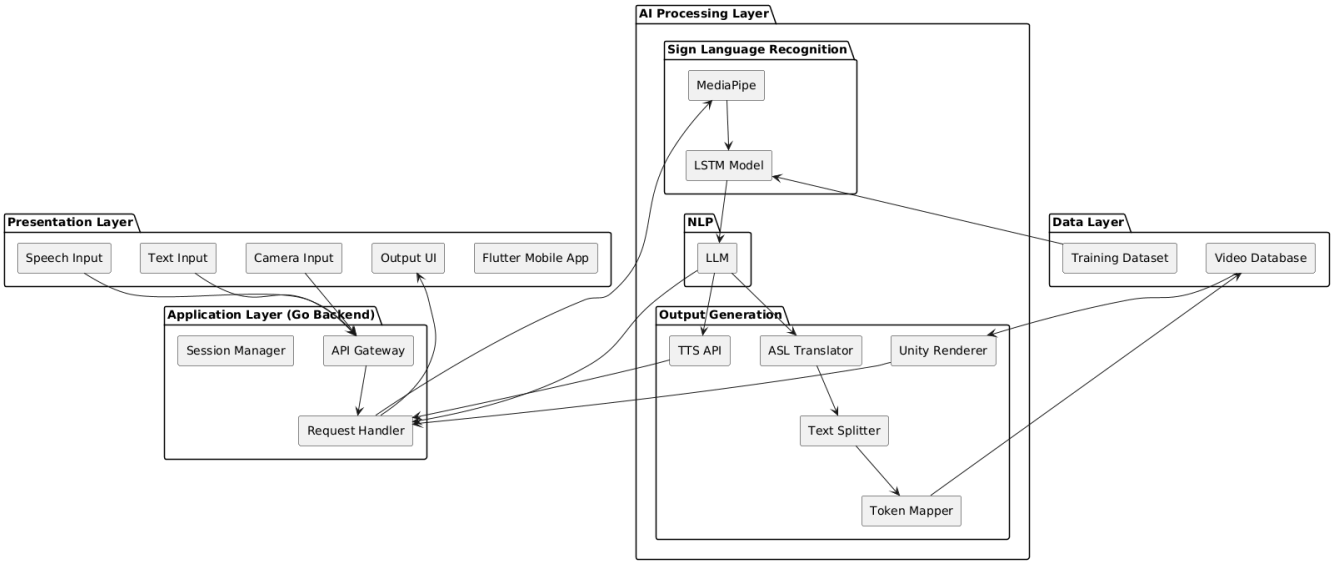

Figure 4.1: System Architecture Diagram of TermTem System

The architecture follows a client-server model consisting of three main layers:

- **Frontend (Mobile Application - Flutter)**
  Handles user interaction including text input, speech input, and real-

time video capture for sign language recognition. It also renders outputs in text, speech, and sign language animation.

- **Backend Services (Go-based API Server)**
  Acts as the central orchestrator, managing communication between frontend and AI components, handling request routing, and coordinating data processing pipelines.
- **AI Processing Pipeline**
  A modular pipeline responsible for transforming user input into meaningful responses. It consists of:
  - MediaPipe for extracting skeletal joint coordinates from video
  - LSTM-based model for sign language recognition
  - Large Language Model (LLM) for generating responses
  - Output modules including Text-to-Speech and Sign Language Animation Generation

The system supports multi-modal input (text, speech, video) and multi-modal output (text, speech, sign language video), making it accessible and flexible for users with different communication needs.

4.2 Software Design

The system runtime behavior is centered around processing user input through a multi-stage pipeline and generating appropriate responses based on the selected output modality.

One of the most critical use cases is real-time sign language translation and response generation, which involves multiple interacting components.

The process begins when the user provides input via real-time video. The frontend captures the video stream and sends it to the backend. The backend forwards the video frames to MediaPipe, which extracts skeletal joint coordinates representing hand movements.

These coordinates are then processed by an LSTM-based model trained for Thai Sign Language recognition, producing a corresponding textual representation.

The recognized text is forwarded to the LLM, which generates a context-aware response. Depending on the selected output mode, the response follows one of three paths:

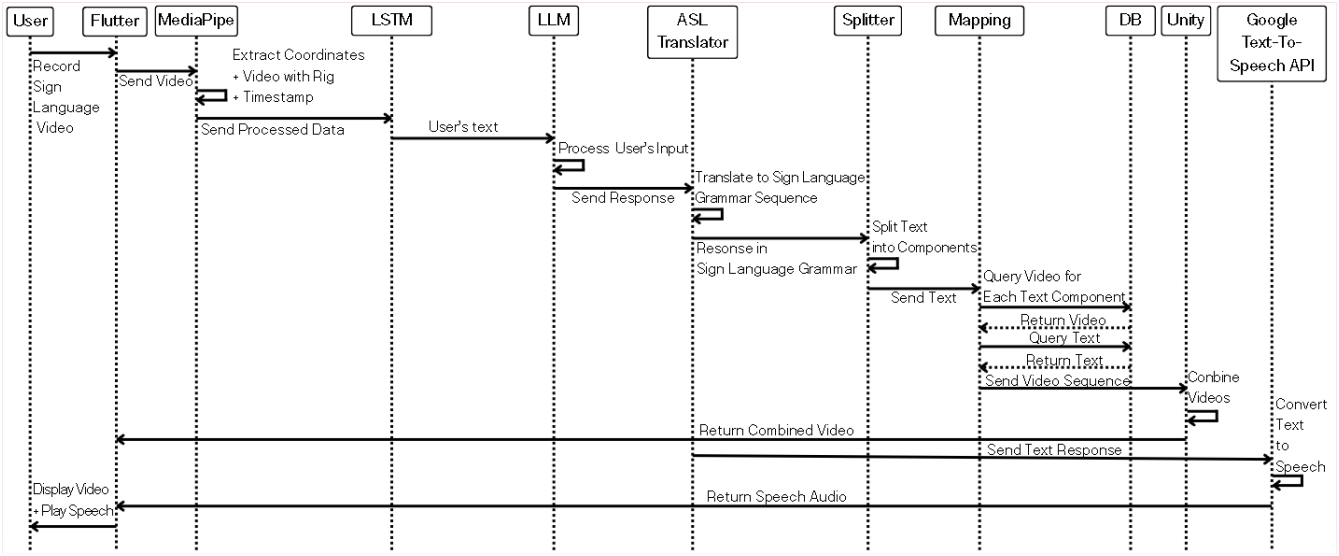

Figure 4.2: Sequence Diagram of TermTem System

1. **Text Response**
   The generated text is sent directly back to the frontend.

2. **Speech Response**
   The text is passed to the Google Text-to-Speech API to generate audio output.

3. **Sign Language Video Response**
   The text is transformed into sign language grammar using an ASL Translator module. The processed sentence is split into tokens, which are mapped to corresponding sign language video clips stored in a database. These clips are sequenced and rendered into a continuous animation using Unity before being sent back to the frontend.

This design ensures modularity and extensibility, allowing future improvements in individual components without affecting the entire system.

4.3 AI and Data Design

4.3.1 Data Sources and Collection

The system utilizes a combination of public, scraped, and self-collected datasets to train and improve the performance of the sign language recognition model.

1. **Public Dataset (Kaggle)**
   The primary dataset consists of over 1,000 Thai sign language videos obtained from Kaggle. These videos are sourced from news agencies and have been pre-labeled and annotated with gesture information. Due to the availability of annotations, minimal preprocessing is required.
2. **Web-Scraped Dataset**
   Additional data is collected from the Deaf Thai Association database. This dataset contains isolated sign language word videos. Since the data is not in a model-ready format, each video is processed using MediaPipe to extract joint coordinate sequences before being used for training.
3. **Self-Collected Dataset (Future Work)**
   If the model performance is insufficient, additional data will be collected manually. This involves recording sign language videos and processing them through MediaPipe to generate training features.

Data Preprocessing includes:

- Video normalization (frame resizing and frame rate alignment)
- Keypoint extraction using MediaPipe
- Sequence padding and truncation for LSTM input
- Label encoding for classification

4.3.2 Model Design

The core AI model for sign language recognition is based on a Long Short-Term Memory (LSTM) architecture.

LSTM is chosen due to its effectiveness in handling sequential temporal data, which is essential for interpreting dynamic hand gestures in sign language. It can capture long-term dependencies between frames, enabling more accurate gesture recognition.

The model pipeline is structured as follows:

- **Input:** Sequence of joint coordinates extracted from MediaPipe
- **Model:** LSTM layers for temporal feature extraction
- **Output:** Predicted text label representing the recognized sign

Model Architecture:

- Input shape: (T, N) where T is the number of frames and N is the number of keypoint features
- LSTM Layer (128 units, return sequences=True)
- Dropout Layer (rate=0.2)
- LSTM Layer (64 units, return sequences=False)
- Dense Layer (64 units, ReLU activation)
- Dropout Layer (rate=0.2)
- Output Dense Layer (Softmax activation)

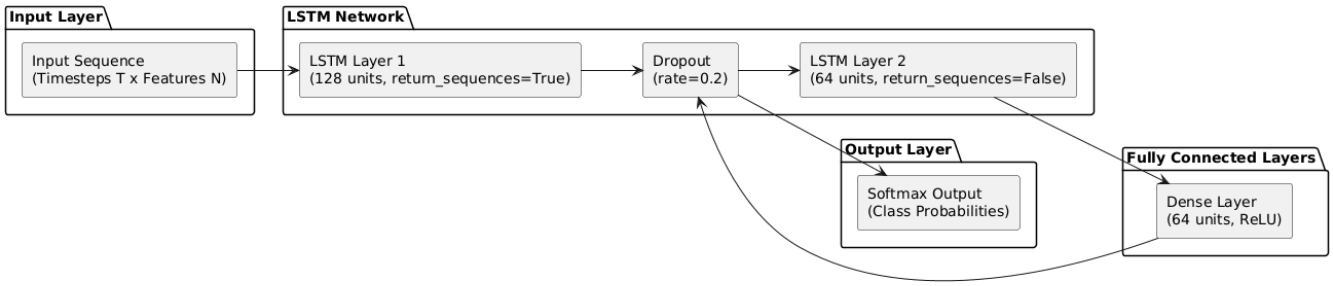

Figure 4.3: Architecture Diagram of Long Short-Term Model

For response generation, a Large Language Model (LLM) is integrated via API to generate context-aware conversational responses.

For sign language output generation, the system uses:

- ASL Translator for grammar transformation
- Text Splitter for tokenization
- Token-to-video mapping using a database
- Unity engine for animation rendering

Training Configuration:

- Loss Function: Categorical Crossentropy
- Optimizer: Adam
- Evaluation Metric: Accuracy

4.3.3 Evaluation

The system performance is evaluated using both quantitative and qualitative metrics:

1. **Recognition Accuracy**
   Measures the correctness of the sign language recognition model using metrics such as accuracy, precision, recall, and F1-score.
2. **System Latency**
   Measures the response time from user input to system output, ensuring near real-time performance.
3. **User Usability Evaluation**
   Conducted through surveys and interviews with users with communication impairments to evaluate usability, satisfaction, and overall user experience.

Chapter 5

Software Development

5.1 Software Development Methodology

The development of the TermTem system follows an **Agile Software Development Methodology**, specifically adopting an iterative and incremental approach. This methodology is chosen due to the experimental nature of AI-based systems, where continuous improvement, testing, and refinement are required.

The development process is divided into multiple short iterations (sprints), each focusing on specific features such as sign language recognition, chatbot integration, and animation generation. At the end of each iteration, a working prototype is produced and evaluated.

Key characteristics of the development methodology include:

- **Iterative Development:** Features are developed in small increments, allowing frequent testing and validation.
- **Continuous Integration:** System components such as the frontend, backend, and AI modules are integrated regularly to ensure compatibility.
- **User-Centered Design:** Feedback from target users, particularly individuals with hearing and speech impairments, is incorporated into the development process to improve usability.
- **Rapid Prototyping:** Early prototypes are built to validate key functionalities such as real-time sign language recognition and response generation.

This methodology enables flexibility in adapting to challenges such as dataset limitations, model performance issues, and real-time system constraints.

5.2 Technology Stack

The TermTem system is built using a combination of modern technologies that support mobile development, backend services, artificial intelligence, and real-time processing.

- **Frontend:**
  - Flutter: Used to develop a cross-platform mobile application with support for real-time camera input, audio processing, and interactive user interfaces.
- **Backend:**
  - Go (Golang): Used to implement a high-performance backend server responsible for handling API requests, managing sessions, and coordinating AI processing pipelines.
- **Artificial Intelligence and Machine Learning:**
  - MediaPipe: Used for extracting skeletal joint coordinates from real-time video input.
  - LSTM Model: Used for recognizing Thai Sign Language gestures based on sequential keypoint data.
  - Large Language Model (LLM): Used to generate context-aware conversational responses.
- **Speech Processing:**
  - Google Text-to-Speech API: Used to convert generated text responses into speech output.
- **Animation and Rendering:**
  - Unity Engine: Used to generate sign language animations by combining sequences of gesture videos.
- **Database:**
  - Video Database: Stores pre-recorded sign language gesture clips mapped to specific tokens.

This technology stack is selected to ensure scalability, real-time performance, and support for multi-modal interaction.

5.3 Coding Standards

To ensure code quality, maintainability, and collaboration efficiency, the TermTem project follows a set of standardized coding practices across all components.

General Principles

- Code should be clean, readable, and well-structured.
- Meaningful and descriptive naming conventions must be used for variables, functions, and classes.
- Each function or module should have a single responsibility.
- Comments should be used to explain complex logic but should not replace clear code.

Frontend (Flutter/Dart)

- Follow **CamelCase** for class names and **lowerCamelCase** for variables and methods.
- Use widget-based modular design for UI components.
- Separate UI logic and business logic where possible.
- Maintain consistent file structure (e.g., screens, widgets, services).

Backend (Go)

- Follow official Go formatting standards using `gofmt`.
- Use short, clear function names and avoid unnecessary complexity.
- Organize code into packages such as handlers, services, and models.
- Handle errors explicitly and consistently.

AI and Machine Learning

- Maintain clear separation between data preprocessing, model training, and inference code.
- Use consistent naming for datasets, features, and labels.
- Document model configurations, hyperparameters, and training procedures.

Version Control

- Use Git for version control.
- Follow a consistent commit message format (e.g., feature, fix, improvement).
- Use separate branches for feature development and merge through pull requests.

5.4 Schedule

<TIP: Show the schedule of your project. Use a Gantt chart. The items should agree with features you listed in Chapter 1. />

5.5 Progress Tracking Report

<TIP: Show that you have been working on this project overtime. It can be in the form of a burndown chart or a contribution graph from GitHub./>

Chapter 6

Deliverables

6.1 Software Solution

<TIP: Share a link to your Github repository. Showcase screenshots of the application and briefly describe each page here. />

6.2 Test Report

<TIP: Describe how you test your project. Place a test report here. If you use continuous integration and deployment (CI/CD) tools, describe your CI/CD method here. />

Chapter 7

Conclusion and Discussion

7.1 Summary of Achievements

<TIP: Summarize what your project accomplished. Refer back to your problem statement from Chapter 1 and explain the degree to which your solution addressed it. Highlight the most significant features or results delivered. />

7.2 Known Limitations

<TIP: Describe the current limitations of your solution honestly. These may include technical constraints (e.g., scalability, performance), scope limitations (features not yet implemented), data limitations (e.g., small training sets, biased data), or usability issues discovered during testing. />

7.3 Future Work

<TIP: Propose concrete improvements or extensions that could be made to this project given more time or resources. These should follow logically from your limitations. For example, if your model was trained on limited data, future work might include collecting a larger dataset. Avoid vague suggestions like "improve the UI"; instead, be specific about what would be changed and why. />

Reference

Appendix A

Appendix A: Example

<TIP: Put additional or supplementary information/data/figures in appendices. />

Appendix B

Appendix B: About LaTeX

LaTeX (stylized as LaTeX) is a software system for typesetting documents. LaTeX markup describes the content and layout of the document, as opposed to the formatted text found in WYSIWYG word processors like Google Docs, LibreOffice Writer, and Microsoft Word. The writer uses markup tagging conventions to define the general structure of a document, to stylize text throughout a document (such as bold and italics), and to add citations and cross-references.

LaTeX is widely used in academia for the communication and publication of scientific documents and technical note-taking in many fields, owing partially to its support for complex mathematical notation. It also has a prominent role in the preparation and publication of books and articles that contain complex multilingual materials, such as Arabic and Greek.

Overleaf has also provided a 30-minute guide on how you can get started on using LaTeX. [**?**]